Generative Modelling with Stochastic Differential Equations

Exercises, week 4

Convergence of diffusion models

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5 Convergence of diffusion models

Setting. Throughout, q is a probability distribution on $\mathbb{R}^n$, $m_2^2 := \mathbb{E}_{X\sim q}|X|^2$ (possibly infinite), $\gamma_n = \mathcal{N}(0, \mathbb{I}_n)$, and $\bar{X}_\tau$ is the forward Ornstein–Uhlenbeck (OU) process

$$\mathrm{d}\bar{X}_\tau = -\bar{X}_\tau\,\mathrm{d}\tau + \sqrt{2}\,\mathrm{d}B_\tau, \qquad \bar{X}_0 \sim q,$$

with marginals $q_\tau = \mathrm{Law}(\bar{X}_\tau)$, scores $s_\tau = \nabla \log q_\tau$ (smooth for $\tau > 0$) and $\sigma_\tau^2 = 1 - e^{-2\tau}$. You may use the following facts.

- *Explicit solution.* For each fixed $\tau > 0$, $\bar{X}_\tau = e^{-\tau}\bar{X}_0 + \sigma_\tau Z_\tau$, where $Z_\tau := (\bar{X}_\tau - e^{-\tau}\bar{X}_0)/\sigma_\tau \sim \gamma_n$ is independent of $\bar{X}_0$. The variable Z_τ depends on τ, and as a process the solution is the stochastic convolution $e^{-\tau}\bar{X}_0 + \sqrt{2}\int_0^\tau e^{-(\tau-r)}\,\mathrm{d}B_r$.
- *Markov form.* For $0 \le \tau' \le \tau$, $\bar{X}_\tau = e^{-(\tau-\tau')}\bar{X}_{\tau'} + \sigma_{\tau-\tau'}Z'$ with $Z' \sim \gamma_n$ independent of $(\bar{X}_r)_{r\le\tau'}$.

The sampler. Fix $T > 0$, a reverse-time grid $0 = t_0 < \cdots < t_K \le T$ with steps $h_k = t_{k+1} - t_k$, and a learned score $\hat{s}_\tau \approx s_\tau$. The exponential-integrator sampler is the process $\hat{Y}$ defined on each $[t_k, t_{k+1}]$ by

$$\mathrm{d}\hat{Y}_t = [\hat{Y}_t + 2\hat{s}_{T-t_k}(\hat{Y}_{t_k})]\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t, \qquad \hat{Y}_0 \sim \gamma_n, \tag{5.1}$$

with B a Brownian motion independent of $\hat{Y}_0$. It approximates the reverse process $\overleftarrow{X}_t = \bar{X}_{T-t}$, whose drift is $\overleftarrow{X}_t + 2s_{T-t}(\overleftarrow{X}_t)$. The *Lipschitz theorem* says that if every s_τ is L-Lipschitz with $L \ge 1$, $t_K = T \ge 1$ and $h = \max_k h_k$, then $\mathrm{KL}(q \,\|\, \mathrm{Law}(\hat{Y}_T)) \le (\frac{1}{2}m_2^2 + ne^{-2T})e^{-2T} + 3L^2nhT + 2\varepsilon_{\text{score}}^2$, where $\varepsilon_{\text{score}}^2$ is the integrated score error $\sum_k h_k \mathbb{E}_{q_{T-t_k}}|\hat{s}_{T-t_k} - s_{T-t_k}|^2$.

Exercise 5.1 (Exponential-integrator update)**.** Let $h > 0$ and let $c, y \in \mathbb{R}^n$ be fixed.

(a) Show that the solution of $\mathrm{d}Y_t = (Y_t + c)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t$, $Y_0 = y$, satisfies

$$Y_h = e^h y + (e^h - 1)c + \sqrt{2}\int_0^h e^{h-r}\,\mathrm{d}B_r \sim \mathcal{N}(e^h y + (e^h - 1)c,\ (e^{2h} - 1)\mathbb{I}_n).$$

Hint: apply Itô's formula to $e^{-t}Y_t$.

(b) Deduce that the scheme (5.1) satisfies

$$\hat{Y}_{t_{k+1}} = e^{h_k}\hat{Y}_{t_k} + 2(e^{h_k} - 1)\,\hat{s}_{T-t_k}(\hat{Y}_{t_k}) + \sqrt{e^{2h_k} - 1}\,\xi_k,$$

where $\xi_0, \ldots, \xi_{K-1}$ are i.i.d. γ_n and independent of $\hat{Y}_0$.

(c) Show that the Euler–Maruyama step $\hat{Y}_{t_k} + h_k(\hat{Y}_{t_k} + 2\hat{s}_{T-t_k}(\hat{Y}_{t_k})) + \sqrt{2h_k}\,\xi_k$ agrees with the update in (b) up to $O(h_k^2)$ in the drift and up to a relative $O(h_k)$ in the noise standard deviation.

Exercise 5.2 (Movement bound)**.** (a) Show that if $0 \le \tau' \le \tau$ and $u = \tau - \tau'$, then $\mathbb{E}|\bar{X}_\tau - \bar{X}_{\tau'}|^2 \le m_2^2 u^2 + 3nu$. *Hint:* use the Markov form of the OU process and the moment bound $\mathbb{E}_{q_\tau}|X|^2 \le n \vee m_2^2$ of Exercise 5.3(a).

(b) What bound holds for large u? For which u is the bound of (a) useful?

(c) Let $q = \gamma_n$. Compute $\mathbb{E}|\bar{X}_\tau - \bar{X}_{\tau'}|^2$ exactly and show that it is $\sim 2nu$ as $u \to 0$. Deduce that, for every L-Lipschitz $f : \mathbb{R}^n \to \mathbb{R}^n$, $\mathbb{E}|f(\bar{X}_\tau) - f(\bar{X}_{\tau'})|^2 \lesssim L^2 nu$ for small u, and explain why the discretisation error in the Lipschitz theorem recalled above is of order $L^2 nhT$ in KL, that is $L\sqrt{nhT}$ in total variation, rather than $L^2 h^2 T$.

Exercise 5.3 (Moment and score bounds)**.** (a) Show that $\mathbb{E}_{q_\tau}|X|^2 = e^{-2\tau} m_2^2 + \sigma_\tau^2 n \le n \vee m_2^2$.

(b) Let p be a positive C^1 probability density on $\mathbb{R}^n$ such that $G := \nabla \log p$ is L-Lipschitz. Show that for every nonnegative $\psi \in C_c^1(\mathbb{R}^n)$,

$$\int \langle G, \nabla \psi \rangle \, \mathrm{d}x \le nL \int \psi \, \mathrm{d}x.$$

Hint: write $\partial_i \psi$ as a limit of difference quotients and move the shift onto G.

(c) Deduce that $\mathbb{E}_p|\nabla \log p|^2 \le nL$, and hence that $\mathbb{E}_{q_\tau}|s_\tau|^2 \le nL$ whenever s_τ is L-Lipschitz. *Hint:* take $\psi = \varphi_R \, p$ with a cutoff φ_R.

(d) Explain why (b)–(c) are a rigorous version of the formal computation $\mathbb{E}_p|\nabla \log p|^2 = -\mathbb{E}_p[\Delta \log p] \le nL$.

(e) Show that equality holds in (c) for $p = \mathcal{N}(0, L^{-1}\mathbb{I}_n)$.

Exercise 5.4 (Two symmetric mixtures)**.** Let $n \ge 2$ and $\mathbf{1}_n = (1, \dots, 1)^\top$.

(a) Let $q = \frac{1}{2}\mathcal{N}(\mathbf{1}_n, \mathbb{I}_n) + \frac{1}{2}\mathcal{N}(-\mathbf{1}_n, \mathbb{I}_n)$. Show that $q_\tau = \frac{1}{2}\mathcal{N}(e^{-\tau}\mathbf{1}_n, \mathbb{I}_n) + \frac{1}{2}\mathcal{N}(-e^{-\tau}\mathbf{1}_n, \mathbb{I}_n)$, compute s_τ and $\nabla^2 \log q_\tau$, and show that $\sup_{\tau \ge 0} \sup_x \|\nabla^2 \log q_\tau(x)\|_{\mathrm{op}} = n - 1$ when $n \ge 2$. Compute $\mathrm{Cov}(q)$ and $\mathbb{E}|\bar{X}_0|^2$.

(b) Let $q = \frac{1}{2}\delta_{\mathbf{1}_n} + \frac{1}{2}\delta_{-\mathbf{1}_n}$. Compute q_τ, s_τ, $\nabla^2 \log q_\tau$, the posterior mean m_τ and the posterior covariance Σ_τ, and check them against the posterior formulas $s_\tau(x) = -\sigma_\tau^{-2} x + e^{-\tau}\sigma_\tau^{-2} m_\tau(x)$ and $\nabla^2 \log q_\tau(x) = -\sigma_\tau^{-2}\mathbb{I}_n + e^{-2\tau}\sigma_\tau^{-4}\Sigma_\tau(x)$. Show that the Lipschitz constant L_τ of s_τ satisfies $ne^{-2\tau}\sigma_\tau^{-4} - \sigma_\tau^{-2} \le L_\tau \le ne^{-2\tau}\sigma_\tau^{-4} + \sigma_\tau^{-2}$.

(c) Use the Lipschitz convergence theorem to estimate the step count for (a). Explain why it does not apply to (b).

Exercise 5.5 (Gaussian score bounds)**.** Let $q = \mathcal{N}(0, v\mathbb{I}_n)$ with $v > 0$. Show that s_τ is L_τ-Lipschitz with $L_\tau = 1/(e^{-2\tau}v + \sigma_\tau^2)$. Check that both bounds of Lemma 5.7 are attained.

Exercise 5.6 (Early stopping)**.** (a) Show that $\mathcal{W}_2(q_\delta, q)^2 \le (1 - e^{-\delta})^2 m_2^2 + n\sigma_\delta^2$, and that the rate $\sqrt{n\delta}$ cannot be improved in general, by computing $\mathcal{W}_2(q_\delta, q)$ for $q = \delta_0$.

(b) Assume $m_2^2 = n$. Given $\varepsilon' > 0$, how small must δ be to guarantee $\mathcal{W}_2(q_\delta, q) \le \varepsilon'$?

Exercise 5.7 (De Bruijn identity and entropy decay)**.** Assume that q has a smooth positive density and that all densities below are smooth enough and decay fast enough to justify differentiation under the integral sign and integration by parts. Let $f_\tau = q_\tau/\gamma_n$ and let $\mathrm{FI}(q_\tau\|\gamma_n) = \int |\nabla \log f_\tau|^2 \, \mathrm{d}q_\tau$ be the relative Fisher information.

(a) Show that the Fokker–Planck equation of the OU process, $\partial_\tau q_\tau = \nabla \cdot (x q_\tau) + \Delta q_\tau$, can be written $\partial_\tau q_\tau = \nabla \cdot (q_\tau \nabla \log f_\tau)$.

(b) Prove the de Bruijn identity $\frac{\mathrm{d}}{\mathrm{d}\tau} \mathrm{KL}(q_\tau\|\gamma_n) = -\mathrm{FI}(q_\tau\|\gamma_n)$.

(c) Using the Gaussian log-Sobolev inequality $\mathrm{KL}(\mu\|\gamma_n) \le \frac{1}{2}\,\mathrm{FI}(\mu\|\gamma_n)$ (valid for every μ with smooth positive density), deduce $\mathrm{KL}(q_T\|\gamma_n) \le e^{-2T}\,\mathrm{KL}(q\|\gamma_n)$, and hence the initialisation bound $\mathrm{TV}(\gamma_n, q_T) \le \sqrt{\mathrm{KL}(q\|\gamma_n)/2}\,e^{-T}$ (Pinsker's inequality: $\mathrm{TV}(\mu,\nu) \le \sqrt{\mathrm{KL}(\mu\|\nu)/2}$). Compare with the bound $\mathrm{KL}(q_T\|\gamma_n) \le \frac{1}{2}(e^{-2T}(m_2^2 - n) - n\log\sigma_T^2)$, valid whenever $m_2 < \infty$. Which one applies when $\mathrm{KL}(q\|\gamma_n) = \infty$?

(d) Check (b) and (c) for $q = \mathcal{N}(m, \mathbb{I}_n)$ and show that the rate e^{-2T} in (c) is attained.

Exercise 5.8 (Learning the score of a Gaussian)**.** Let $q = \mathcal{N}(\mu, \mathbb{I}_n)$ with unknown $\mu \in \mathbb{R}^n$, and $X_1, \dots, X_N \overset{\text{i.i.d.}}{\sim} q$. For an estimator $\hat{s}_\tau$ built from the sample, let $R_\tau(\hat{s}_\tau) := \mathbb{E}_{X\sim q_\tau}|\hat{s}_\tau(X) - s_\tau(X)|^2$, with X independent of the sample.

(a) Compute q_τ and s_τ.

(b) For the plug-in estimator $\hat{s}_\tau(x) = -(x - e^{-\tau}\hat{\mu})$ with $\hat{\mu} = \frac{1}{N}\sum_i X_i$, show $\mathbb{E}R_\tau(\hat{s}_\tau) = e^{-2\tau}n/N$.

(c) Show that every estimator $\hat{s}_\tau$ (any measurable function of the sample and of the evaluation point) satisfies $\sup_\mu \mathbb{E}R_\tau(\hat{s}_\tau) \ge e^{-2\tau}n/(N+1)$. *Hint:* put a prior $\mu \sim \mathcal{N}(0, \lambda\mathbb{I}_n)$ and let $\lambda \to \infty$.

(d) Compute $\int_0^\infty \mathbb{E}R_\tau(\hat{s}_\tau)\,\mathrm{d}\tau$ for the plug-in estimator. Show that $\mathbb{E}\sum_k h_k R_{T-t_k}(\hat{s}) \le n/(2N)$ for any grid ending at or before T.